A3. Decision Table Testing

Decision Table Testing is used when the system has multiple conditions that can affect the final result.

Here, `registration_decision()` checks five conditions:

1. Does the student have outstanding dues?
2. Is the student already registered?
3. Are all prerequisites completed?
4. Can the student take additional credits?
5. Is a seat available?

According to FR-09, the system must follow a specific priority order. It returns the first applicable outcome.

1. Conditions and actions

Condition	True means
C1: Dues greater than zero?	Registration blocked
C2: Already registered?	Duplicate registration
C3: Prerequisites satisfied?	Student is eligible
C4: Credit limit satisfied?	Student can take the credits
C5: Seat available?	Course has free seats

2. Decision table

Here, – means the condition does not need to be checked because an earlier condition has already determined the result.

Conditions / Rules	R1	R2	R3	R4	R5	R6
C1: Dues > 0	T	F	F	F	F	F
C2: Already registered	–	T	F	F	F	F
C3: Prerequisites satisfied	–	–	F	T	T	T
C4: Credit limit satisfied	–	–	–	F	T	T
C5: Seat available	–	–	–	–	T	F

Expected result	DUES_BLOCK	ALREADY_REGISTERED	PREREQUISITES_MISSING	CREDIT_LIMIT	REGISTERED	WAITLISTED

This table covers all six possible decision outcomes.

3. Executable decision-table test cases

Test ID	Conditions	Expected output	Purpose
TC36	Dues = 1	DUES_BLOCK	Test dues priority
TC37	Dues = 0, already registered	ALREADY_REGISTERED	Test duplicate registration
TC38	No dues, not registered, prerequisites missing	PREREQUISITES_MISSING	Test prerequisite rejection
TC39	Prerequisites satisfied, credits exceed 18	CREDIT_LIMIT	Test credit rejection
TC40	All eligible, seats available	REGISTERED	Successful registration
TC41	All eligible, course full	WAITLISTED	Test waiting list

4. Priority testing

We should also test combinations where multiple conditions are true simultaneously.

Test ID	Dues	Already registered	Prerequisites	Credit eligible	Seat available	Expected result
TC42	Yes	Yes	No	No	No	DUES_BLOCK
TC43	No	Yes	No	No	No	ALREADY_REGISTERED
TC44	No	No	No	No	No	PREREQUISITES_MISSING
TC45	No	No	Yes	No	No	CREDIT_LIMIT

Why are these important?

For example, in TC42, the student has outstanding dues and is already registered.

The system must return `DUES_BLOCK`, not `ALREADY_REGISTERED`, because dues have higher priority.

Similarly, TC43 verifies that duplicate registration has priority over missing prerequisites.

These cases verify that the system follows the exact priority specified in FR-09.